

A nonnumerical multiplier in an odd-order difference set

Disproof of Conjecture 00000008412

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Abstract

We exhibit a $(27, 13, 6)$ difference set in $(\mathbb{Z}/3\mathbb{Z})^3$ preserved by the coordinate cycle $(a, b, c) \mapsto (b, c, a)$. This is a group automorphism and a nonnumerical multiplier. The ambient group has no even-order subgroup or subgroup quotient, and no index-two subgroup. Thus nonnumerical multipliers are not confined to the asserted $(16, 6, 2)$ exception or its index-two extension. All finite difference counts and the automorphism are verified in Lean.

Definitions and scope

For an additive finite group G , a (v, k, λ) difference set is a k -element subset D with $|G| = v$ such that each nonzero $g \in G$ occurs exactly λ times as $x - y$ with $(x, y) \in D^2$. A multiplier is an automorphism σ of G such that $\sigma(D) = D + t$ for some $t \in G$. In an abelian group a multiplier is numerical if $\sigma(x) = nx$ for an integer n . These are the usual abelian conventions; see Ó Catháin, Section 3.1.

We refute the only-if direction of the conjecture's second clause: the restriction of nonnumerical multipliers to the stated even-order exception. The source does not separately define “coexists.” The exclusion proved here covers an exception in the ambient group, a subgroup, a quotient of a subgroup, or an index-two extension of a group of order 16. Indeed, no even-order group section is possible. We do not assign a meaning involving unrelated groups to that word. No claim about the two other clauses is needed.

The difference set

Let $G = (\mathbb{Z}/3\mathbb{Z})^3$, with coordinatewise addition and subtraction, and let D be the following set of thirteen triples:

$$D = \{(0, 0, 1), (0, 1, 0), (0, 1, 1), (0, 1, 2), \\ (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1), \\ (1, 2, 0), (1, 2, 2), (2, 0, 1), (2, 1, 2), (2, 2, 1)\}.$$

Its elements are distinct. For $g \in G$ write $N(g) = \#\{(x, y) \in D^2 : x - y = g\}$. Direct subtraction modulo 3 gives the full table below. The row is the first coordinate of g ; the column is its last two coordinates.

	00	01	02	10	11	12	20	21	22
0	13	6	6	6	6	6	6	6	6
1	6	6	6	6	6	6	6	6	6
2	6	6	6	6	6	6	6	6	6

For example, for $g = (0, 0, 1)$ the six possible first entries x are $(0, 1, 0)$, $(0, 1, 1)$, $(0, 1, 2)$, $(1, 0, 1)$, $(1, 1, 1)$, and $(1, 2, 0)$; each has $y = x - g \in D$. Each table entry can equivalently be obtained by testing $x - g \in D$ for the thirteen displayed x . Thus D is a nontrivial $(27, 13, 6)$ difference set.

A multiplier that is not numerical

Define $T(a, b, c) = (b, c, a)$. This map is additive, has inverse $T^{-1}(a, b, c) = (c, a, b)$, and satisfies $T^3 = 1$. The displayed set D is a union of the following T -orbits:

$$\begin{aligned} &\{(1, 1, 1)\}, \\ &\{(0, 0, 1), (0, 1, 0), (1, 0, 0)\}, \\ &\{(0, 1, 1), (1, 1, 0), (1, 0, 1)\}, \\ &\{(0, 1, 2), (1, 2, 0), (2, 0, 1)\}, \\ &\{(1, 2, 2), (2, 2, 1), (2, 1, 2)\}. \end{aligned}$$

Consequently $T(D) = D$, so T is a multiplier with translation 0. It is not numerical: for every integer n ,

$$T(1, 0, 0) = (0, 0, 1) \neq (n \bmod 3, 0, 0) = n(1, 0, 0).$$

This excludes every integer scalar map, and therefore also all numerical automorphisms with the usual coprimality condition.

Excluding the proposed exception

The ambient group has order 27, not 16 or 32. More strongly, if $S \leq G$ and $K \leq S$, Lagrange's theorem gives

$$|S/K| \mid |S| \mid 27.$$

Hence every such group section has odd order. In particular neither an order-16 group supporting a $(16, 6, 2)$ difference set nor its order-32 index-two extension can occur in any such section. Also $[G : S] \mid 27$, so G has no subgroup of index two.

Theorem 1. *A nontrivial difference set can have a nonnumerical multiplier in an odd-order group having no even-order group section. Thus the claimed restriction to the $(16, 6, 2)$ exception or its index-two extension in Conjecture 00000008412 is false.*

Proof. The difference set $D \subseteq (\mathbb{Z}/3\mathbb{Z})^3$ and its multiplier T have exactly these properties. $\square$

Formal correspondence and reproduction

Lean uses the actual product additive group $\mathbb{ZMod}\ 3 \times \mathbb{ZMod}\ 3 \times \mathbb{ZMod}\ 3$. The predicate `IsDifferenceSet` counts ordered pairs with their actual group difference. The full $(27, 13, 6)$ property is kernel checked. The map `cycle` is a genuine additive equivalence with an explicit inverse; its additivity, its cube, and preservation of D are proved. The multiplier predicate allows translations. The theorem `cycle_not_numerical` quantifies over all integers, not just selected residues.

The theorem `no_even_section` uses Mathlib's Lagrange theorem for arbitrary subgroups and subgroup quotients. Together with the index-two exclusion it proves the structural facts in the final `counterexample`. These are group properties, not tests on a list of guessed subgroup orders.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. In `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The submitted project is rebuilt in a fresh directory. There are no proof placeholders, custom axioms, or external computational oracles. The accompanying `verify.py` reproduces the original orbit-union search and all finite counts using Python’s standard library; Lean does not trust its output. Run `python verify.py` for that check and `tectonic main.tex` to regenerate the PDF. The author performed a separate adversarial self-review; no independent reviewer or subagent was used.

References

1. The Last Math Competition, Conjecture 00000008412. Its exact bilingual source and upstream provenance are included in the submission.
2. P. Ó Catháin, *Inequivalence of difference sets: On a remark of Baumert*, Electronic Journal of Combinatorics 20(1) (2013), P38, Section 3.1. Publisher’s full text.